

Project 24

Capital Radiology - Online System for Magnetic Resonance Imaging (MRI) Lab

Objectives of the System

This project plays a vital role in enhancing the online services being available worldwide. The patients obtain a MRI scan to diagnose and treat medical conditions. Magnetic Resonance Imaging (MRI) is non-invasive and painless medical tests that allow physicians to better evaluate parts of the body and certain diseases that may not be assessed adequately with other imaging methods such as x-ray, ultrasound or computer tomography.

This project contains six modules:

- Secure Authentication Process
- Web Portal Implementation
 - Storage of MRI image
 - Designing the Report
- Leakage Record Prevention
- Doctor's Prescription to the Patients

EXISTING SYSTEM

Generally, reports are printed, and the patient travels for consultation from the doctor for a report. Patients are struggle by the following process

- Patient is waiting in the queue for the consultant from the doctor.
- Lot of cost expenses.
- Lot of time duration and man power.

PROPOSED SYSTEM

The patients or an admin send the MRI image to this portal so that the doctors can view over it and send back their report based on it. The report can

be in a predefined format as designed like a user interface, such as by gathering the Patient status. The doctors should be available online for sending the reports and guidance back to the patient. The project reduces this effort and time involved by using online services via internet. The aim is to design an online service or a WEB PORTAL where doctors can subscribe themselves to provide the reports to patients.

MODULES

1. User Management
2. Appointment Scheduling
3. Patient Profile Management
4. Order and Referral Management
5. MRI Procedure Management
6. Image Storage and Retrieval
7. Reporting
8. Billing and Payment
9. Notifications and Communication
10. Results Delivery and Patient Portal
11. Staff Management
12. Equipment Management
13. Inventory Management
14. Admin Dashboard and Management
15. Reporting and Analytics
16. Security and Audit
17. Content Management

1. User Management Module

- Patient registration and login
- Staff and radiologist account management
- Admin account management
- Password reset and account recovery
- Role-based access control (Patient, Technician, Radiologist, Admin)

2. Appointment Scheduling Module

- Online booking for MRI scans
- Select preferred date, time slot, and location
- View available slots in calendar
- Staff-side scheduling and approval
- Automated email/SMS confirmations and reminders

3. Patient Profile Management Module

- Store demographic details (name, DOB, contact info)
- Manage medical history and contraindications
- Upload referral documents
- View appointment history
- Manage emergency contact information

4. Order and Referral Management Module □

Upload physician referral orders

- View and manage pending referrals
- Assign referrals to specific appointments
- Notify referring doctors of scheduled scans

5. MRI Procedure Management Module

- Record scan details (body part, protocol, settings)
- Assign technician and radiologist
- Track scan status (Scheduled, In Progress, Completed, Reported)
- Log equipment used and scan duration

6. Image Storage and Retrieval Module

- Store MRI images in secure PACS or database
- Upload DICOM images from scanner
- Link images to patient records and appointment
- Enable radiologists to view and annotate images
- Support exporting images for sharing

7. Reporting Module

- Allow radiologists to create structured MRI reports
- Store reports securely and link to appointments
- Support template-based reporting
- Approve and sign final reports
- Share reports with referring doctors and patients

8. Billing and Payment Module

- Generate bills based on procedures
- Support multiple payment methods (card, insurance)
- Issue digital receipts
- Track payment history
- Handle insurance claims and approvals

9. Notifications and Communication Module

- Email/SMS notifications for booking confirmations, reminders, and results ready
- Secure messaging between patients and staff
- Notify referring physicians when reports are complete
- Internal messages between technicians, radiologists, and admin

10. Results Delivery and Patient Portal Module

- Patients can log in to view completed MRI reports
- Secure download of DICOM images if permitted
- View billing history and receipts
- Manage future appointments
- Receive and manage notifications

11. Staff Management Module

- Create and manage technician, radiologist, and admin accounts
- Assign roles and permissions
- Manage staff schedules and availability
- Track staff activity logs

12. Equipment Management Module

- Register MRI scanners and related equipment
- Track usage and maintenance schedules
- Log equipment calibration and service history
- Generate maintenance reminders

13. Admin Dashboard and Management Module

- View overall system statistics (appointments, scans performed)
- Manage users and roles
- Oversee appointments and workflow
- Access financial summaries and payment reports
- Configure system settings

14. Reporting and Analytics Module

- Generate daily, weekly, monthly scan volume reports
- Financial and billing summaries
- Staff utilisation and workload reports
- Equipment usage and downtime analysis
- Export reports in PDF or CSV format

15. Security and Audit Module

- Enforce user authentication and access control
- Encrypt patient data at rest and in transit
- Log all user activities for auditing
- Role-based permissions for sensitive data
- Generate audit trails for compliance (HIPAA/GDPR)

16. Content Management Module

- Manage static pages (About Us, Contact, FAQs)
- Upload MRI preparation instructions
- Manage service details and pricing
- Update announcements or alerts on the portal

FUNCTIONAL REQUIREMENTS

FR1. Allow patient registration with email verification.

FR2. Provide secure login and logout for all users.

FR3. Enable password reset and account recovery.

FR4. Let users edit personal profile information.

FR5. Enforce role-based access for patients, technicians, radiologists, and administrators.

FR6. Enable patients to book MRI appointments online.

FR7. Allow selection of preferred date, time slot, and location.

FR8. Display available appointment slots in calendar view.

FR9. Let staff schedule, edit, or cancel appointments.

FR10. Send automated email or SMS confirmations and reminders.

FR11. Store patient demographic details, including name, date of birth, and contact information.

FR12. Maintain medical history and contraindications (e.g., implants, allergies).

FR13. Allow upload of physician referral documents.

FR14. View past and upcoming appointment history.

FR15. Manage emergency contact details.

FR16. Upload and store physician referral orders.

FR17. View and manage pending referrals.

FR18. Assign referrals to scheduled appointments.

FR19. Notify referring doctors when appointments are scheduled.

FR20. Record MRI scan details such as body part, protocol, and settings.

FR21. Assign specific technician and radiologist to each procedure.

FR22. Track scan status (Scheduled, In Progress, Completed, Reported).

FR23. Log equipment used and scan duration.

FR24. Store MRI images securely in PACS or internal database.

FR25. Upload DICOM images from scanners.

FR26. Link images to specific patient records and appointments.

FR27. Allow radiologists to view and annotate images.

FR28. Support image export and sharing with authorised users.

FR29. Enable radiologists to create structured MRI reports.

FR30. Store and link reports to corresponding appointments.

FR31. Provide template-based reporting options.

FR32. Allow electronic signing and finalisation of reports.

FR33. Share completed reports securely with referring doctors and patients.

FR34. Generate itemised bills based on procedures performed.

FR35. Support multiple payment methods, including card and insurance.

FR36. Issue digital receipts for completed payments.

FR37. Track payment history for patients.

FR38. Manage insurance claims and approvals.

FR39. Send email or SMS notifications for appointment confirmations and reminders.

FR40. Notify patients when MRI reports are ready.

FR41. Enable secure messaging between patients and staff.

FR42. Notify referring physicians when reports are complete.

FR43. Allow internal communication between technicians, radiologists, and admins.

FR44. Allow patients to log in to view their completed MRI reports.

FR45. Enable secure download of DICOM images if permitted.

FR46. Let patients view billing history and receipts.

FR47. Provide ability to manage upcoming appointments.

FR48. Enable patients to manage their notification preferences.

FR49. Allow creation and management of staff accounts.

FR50. Assign roles and permissions to staff members.

FR51. Manage staff schedules and availability.

FR52. Track and log staff activity for auditing purposes.

FR53. Register MRI scanners and related equipment.

FR54. Track equipment usage and maintenance schedules.

FR55. Log calibration records and service history.

FR56. Generate reminders for upcoming maintenance.

FR57. Track inventory levels of contrast agents and consumables.

FR58. Record stock in/out transactions.

FR59. Generate alerts for low inventory.

FR60. Store supplier details and contacts.

FR61. Report on inventory usage and wastage.

FR62. Display overall system statistics in an admin dashboard.

FR63. Manage user accounts and roles.

FR64. Oversee all scheduled appointments and workflow.

FR65. View financial summaries and payment reports.

FR66. Configure system settings and preferences.

FR67. Generate daily, weekly, and monthly scan volume reports.

FR68. Produce financial and billing summaries.

FR69. Export reports in PDF or CSV format.

FR70. Enforce secure user authentication and access controls.

FR71. Encrypt patient data at rest and in transit.

FR72. Log all user activities for audit and compliance.

FR73. Enforce role-based permissions for accessing sensitive data.

FR74. Generate detailed audit trails to meet regulatory standards (HIPAA/GDPR).

FR75. Manage static site content such as About Us, Contact, and FAQs pages.

FR76. Upload patient instructions for MRI preparation.

FR77. Update service details and pricing.

FR78. Post announcements or alerts to the patient portal.

NON FUNCTIONAL REQUIREMENTS

1. Performance Requirements

NFR1. Support at least 200 users with response times under 2 seconds for most operations.

NFR2. Handle large image file uploads and downloads (e.g., DICOM) without significant performance degradation.

NFR3. Provide real-time search for appointments, patients, and reports with minimal latency.

NFR4. Enable scalable architecture to handle increased demand over time.

2. Security Requirements

NFR5. Encrypt patient data at rest using industry-standard encryption (e.g., AES-256).

NFR6. Enforce HTTPS for all data transmission to secure data in transit.

NFR7. Use secure password hashing algorithms.

NFR8. Enforce role-based access control with fine-grained permissions.

NFR9. Log all user activity for auditing and compliance.

NFR10. Detect and alert administrators about suspicious or unauthorised access attempts.

3. Availability Requirements

NFR11. Maintain at least 99.5% system uptime excluding planned maintenance.

NFR12. Support automated failover to minimise downtime during server failures.

NFR13. Provide clear maintenance notifications to users in advance.

4. Reliability Requirements

NFR14. Ensure data integrity during all transactions (e.g., booking, image uploads, reporting).

NFR15. Provide automated database backup and restore capabilities.

NFR16. Guarantee that completed transactions are never lost even during partial failures.

5. Usability Requirements

NFR17. Deliver an intuitive, user-friendly interface for patients, technicians, and radiologists.

NFR18. Support responsive design for desktop, tablet, and mobile access.

NFR19. Provide clear validation and error feedback to users.

NFR20. Ensure accessibility compliance with WCAG 2.1 Level AA guidelines where feasible.

6. Compatibility Requirements

NFR21. Support major modern web browsers (Chrome, Firefox, Safari, Edge).

NFR22. Ensure mobile compatibility or dedicated mobile app integration.

NFR23. Use standard formats for medical images (DICOM) and reports (PDF).

7. Scalability Requirements

NFR24. Provide modular architecture to support adding new features without major refactor.

8. Maintainability Requirements

NFR25. Follow consistent coding standards and documentation practices.

NFR26. Use a modular codebase for easier updates and feature enhancements.

NFR27. Log and monitor application errors with detailed diagnostics.

9. Audit and Compliance Requirements

NFR28. Maintain audit logs of critical user activity for at least 12 months.

NFR29. Support export of audit logs for compliance reviews.

NFR30. Ensure data handling complies with relevant regulations (e.g., HIPAA, GDPR).

10. Backup and Recovery Requirements

NFR31. Perform automated daily database backups.

NFR32. Support on-demand manual backups by administrators.

NFR33. Enable system recovery within 4 hours of a critical failure.

12. Legal and Ethical Requirements

NFR34. Display clear Terms of Service and Privacy Policy to all users.

NFR35. Obtain patient consent for data collection and storage.

NFR36. Provide mechanisms for patients to request account deletion and data removal.

NFR37. Ensure ethical handling of sensitive patient health data in line with medical standards.

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.